Problem Set 8: The Physics of the PWR

Due: Friday, March 27, 2026

Instructions

- Show all work, including the equations used and unit conversions.
- **Constants for Water at PWR Operating Conditions (15.5 MPa):**
  - Specific Heat Capacity: $C_p \approx 5.8$ kJ/kg°C
  - Saturation Temperature: $T_{sat} \approx 345$°C
  - Density at 300°C: $\rho \approx 712$ kg/m^3
- **Fuel Properties (UO_2):**
  - Thermal Conductivity: $k_f \approx 3.0$ W/m · K
  - Melting Point: $T_{melt} \approx 2800$°C

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1. **The Ford F-150 Engine (Primary Loop Thermodynamics)**
   A standard Westinghouse 4-Loop PWR produces 3411 MW$_{th}$ of thermal power. The coolant enters the core at $T_{in} = 290$°C and exits at $T_{out} = 325$°C. The system pressure is maintained at 15.5 MPa (2250 psia).
   (a) Calculate the total **Mass Flow Rate** ($\dot{m}$) of the primary coolant through the core in kg/s.
   (b) Calculate the **Margin to Saturation** at the core exit (i.e., how many degrees cooler is the outlet water than the boiling point?).
   (c) Why is this margin necessary? (Think about what happens at the surface of the hot fuel rods versus the bulk water temperature).

2. **The "Ceramic Limit" (Fuel Rod Sizing)**
   In Lecture 26, we derived that the temperature drop across a fuel pellet is given by $\Delta T = \frac{q'''R^2}{4k_f}$, where q''' is the volumetric power density and R is the pellet radius.
   (a) Consider a standard fuel pellet with radius $R = 0.41$ cm. The reactor is operating at a high volumetric power density of $q''' = 350$ MW/m^3. If the surface temperature of the pellet is $T_s = 400$°C, calculate the **Centerline Temperature** (T_{CL}). Is this safe?
   (b) An engineer proposes "simplifying" the design by doubling the rod radius to $R_{new} = 0.82$ cm while maintaining the same economic power density ($q''' = 350$ MW/m^3). Calculate the new centerline temperature.

(c) What would happen to the fuel in scenario (b)?

3. The Transparent Lattice

The standard PWR lattice has a fuel rod pitch of $P = 1.26$ cm.

(a) A fast fission neutron (2 MeV) has a macroscopic scattering cross-section in water of $\Sigma_s \approx 0.15$ cm^{-1}. Calculate its Mean Free Path (λ_{fast}).

(b) A thermal neutron (0.025 eV) has a transport cross-section in water of $\Sigma_{tr} \approx 3.0$ cm^{-1}. Calculate its Mean Free Path ($\lambda_{thermal}$).

(c) Compare both values to the Pitch P. Based on this comparison, explain why we say the fast neutrons "see a homogeneous mist" while thermal neutrons "see a distinct heterogeneous lattice."

4. Reactivity Control with Boron

Instead of using control rods for burnup compensation, PWRs use Soluble Boron (Chemical Shim).

(a) A fresh core has an excess reactivity of $\rho_{excess} = 0.15$ ($15\%\Delta k/k$). If the Differential Boron Worth is $\alpha_B = -8$ pcm/ppm (where 1 pcm $= 10^{-5}\Delta k/k$), what concentration of Boron (in ppm) is required to hold the reactor critical at the Start of Cycle (BOL)?

(b) Safety regulations limit the maximum Boron concentration to ≈ 1500 ppm to prevent a positive Moderator Temperature Coefficient (MTC). Based on your answer in (a), would this core design be permitted?

(c) If the limit is 1500 ppm, what is the maximum excess reactivity (and thus fuel cycle energy) we can load into the core?

5. The "Lung" of the System

The Pressurizer is essential because water expands when heated.

- Total Primary System Volume (excluding pressurizer): $V_{sys} = 300$ m^3.
- Pressurizer Volume: $V_{prz} = 40$ m^3.

During a load rejection transient, the average coolant temperature spikes from 300°C to 310°C.

(a) Given that water density decreases from 712 kg/m^3 (at 300°C) to 696 kg/m^3 (at 310°C), calculate the **increase in volume** (ΔV) of the liquid water in the primary loops.

(b) If the Pressurizer was initially 50% full of liquid, what is the new liquid level percentage? (Assume the surge line allows water to flow freely into the pressurizer).

(c) What would happen to the system pressure if the Pressurizer were full of water (solid) instead of having a steam bubble?